सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/37 **Date:** 16-05-2025

As per distribution list

Subject: 37th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 37th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 27th March 2025 (Thursday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

(Partha Sarathi Das)
Sr. General Manager (CTU)

"सौदामिनी", प्रथम तल, प्लॉट सं.2, सेक्टर-29, गुरुग्राम- 122001 (हरियाणा), दूरभाष: 0124-2822000, सीआईएन: U40100HR2020GOI091857
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Minutes of 37th Consultation Meeting for Evolving Transmission Schemes in Northern Region scheduled to be held on 27.03.2025

A. Review of dedicated Transmission System (Under Applicant Scope) of connectivity granted to M/s SA Renewable Infra Pvt. Ltd. (App. No. 2200000755) at Bikaner-V PS

It was informed that M/s SA Renewable Infra (App. No. 220000755) has been granted connectivity of 100 MW through sharing of dedicated transmission system of M/s Avaada RJ Bikaner (2200000561 – 136 MW) at 220 kV Bikaner-V PS in 33rd CMETS NR meeting held on 05.08.2024. Subsequently, in the 36th CMETS-NR meeting held on 15.01.2025, it was agreed to grant connectivity of 354 MW to M/s Avaada RJBikaner(App. No. 2200001326) at 220 kV Bikaner-V PS. Further, in the 36th CMETS-NR meeting, it was agreed that both the connectivity applications of M/s Avaada (2200000561:136MW & 2200001326: 354MW) may be combined together in terms of common PS with total 490MW connectivity through 220kV D/c line. Considering the total connectivity granted to M/s Avaada RJ Bikaner(490 MW: 136 MW +354 MW), the dedicated transmission system was revised from earlier granted 220 kV S/c line to 220 kV D/c line(suitable to carry minimum 300 MW per circuit at nominal voltage).

Accordingly, in the 36th CMETS NR meeting, it was decided that the connectivity granted to M/s SA renewable shall be reviewed and shall be considered in sharing with other RE projects in Bikaner-V PS in view of the change in dedicated transmission system granted to M/s Avaada RJ Bikaner.

It was informed that, with the total grant of 5830 MW connectivity at Bikaner-V PS, Bikaner-V PS is already closed for fresh grant except enhancement(170 MW margin available at 220 kV bay granted to Juniper(130 MW)). After grant of 490 MW connectivity to Avaada RJ Bikaner, there is margin available of 110 MW in the dedicated transmission system (220 kV D/c line). Therefore, it was informed that M/s SA renewable may retain their connectivity at Bikaner-V PS in sharing with M/s Avaada RJ Bikaner or may shift connectivity in sharing with other RE applicants having margin in their 220kv bay at Bikaner-V PS.

M/s SA Renewable in the meeting & vide mail dated 24.03.2025 informed that their sister concern Iraax International (App. No. 2200000540(150 MW) & 2200000484(100MW)) has also been granted connectivity at 220 kV Bikaner-V PS. Therefore M/s SA Renewable wish to share the connectivity with Iraax international pvt ltd in place of Avaada so as to optimally utilize the ROW and land parcels. Further, M/s SA Renewable also apprised that they are ready to terminate connectivity sharing agreement with Avaada and enter the same with Iraax international. M/s Avaada Energy in the meeting informed that they have no objection to SA Renewable terminating the sharing agreement with Avaada and enter the same with Iraax international.

It was also informed that M/s Iraax International need to consider implementation of DTL with capacity of 350 MW to accommodate SA Renewable. M/s Iraax International agreed for the same. Considering above, it was agreed to revise the connectivity granted to SA Renewable in sharing with M/s Iraax International & revised intimation for connectivity to SA Renewable & M/s Iraax shall be issued accordingly. M/s SA renewable & Iraax noted the same. Further, M/s SA Renewable need to submit sharing agreement with M/s Iraax for sharing of bay & DTL. M/s SA Renewable noted the same

B. Application related matters in Northern Region

C1. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 deferred in the 36th CMETS NR meeting:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200001352	Avaada Energy Private Limited	Barmer distt., Rajasthan	14.10.2024	Generator (Solar)	NHPC LOA	750	31.03.2027	Barmer-II PS
2.	2200001417	Sprng Green Energy 4 Private Limited	Bikaner distt., Rajasthan	29.10.2024	Generator (Solar)	Land BG Route	400	30.06.2029	Bhadla-IV PS
3.	2200001420	Azure Power Fifty One Private Limited	Jodhpur distt., Rajasthan	30.10.2024	Generator (Solar)	Land Route	600	31.12.2029	Bhadla-IV PS
4.	2200001422	Avaada Energy Private Limited	Barmer distt., Rajasthan	30.10.2024	Generator (Solar with ESS)	SJVN LOA	800	30.06.2027	Barmer-II PS

Connectivity at Barmer Complex:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
1.	2200001352	Avaada Energy Private Limited	Barmer distt., Rajasthan	14.10.2024	Generator (Solar)	NHPC LOA	750	31.03.2027	Barmer-II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW
2.	2200001422	Avaada Energy Private Limited	Barmer distt., Rajasthan	30.10.2024	Generator (Solar with ESS)	SJVN LOA	800	30.06.2027	Barmer-II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that, M/s Avaada Energy Private Limited has applied for connectivity for 750 MW & 800 MW at Barmer-II PS vide above applications. The applications were discussed in the 36th CMETS NR meeting held on 15.01.2025 wherein M/s Avaada was asked to clarify regarding the technical feasibility of the projects location considering possible infringement of GIB priority area. Subsequently, M/s Avaada vide mail dated 27.02.2025 clarified that M/s Avaada will be setting up a

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

pooling station for these projects, connected through a 400 kV D/c transmission line to Barmer-III PS & the distance from the pooling station to Barmer-III PS shall be technically feasible and requested CTUIL to process the applications. Accordingly, it was proposed to grant connectivity at Barmer-III PS through 400 kV D/c line for the combined capacity of 1550 MW (750MW+800MW).

Applicant was asked to confirm the scope of 400 kV line bays at Barmer-III PS end. M/s Avaada informed that the 400 kV Bays at Barmer-III PS shall be under ISTS scope. Same was noted.

Further, it was mentioned that M/s Avaada Energy has applied for connectivity of 800 MW(App. No. 2200001422) which is more than the contracted capacity of LoA(590 MW). Regarding the grant of connectivity quantum higher than contracted capacity, there is an ongoing case filed by Axis Energy in Andhra Pradesh high court. Therefore, the connectivity of 800 MW shall be granted to M/s Avaada Energy(App. No.2200001422) and the same shall be subject to outcome of the decision of Andhra Pradesh High Court in this regard. M/s Avaada noted the same.

It was informed that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Avaada Energy Private Limited has requested for Start Date of Connectivity from 31.03.2027(App. No. 2200001352 - 750 MW) & 30.06.2027(App. No. 2200001422 - 800 MW). However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity shall be 30.09.2030 (interim). M/s Avaada noted the same.

Details of Transmission system for connectivity of M/s Avaada Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). Common Pooling station for Avaada Energy Private Limited RE Power Projects (App. No. 22000001352 – 750 MW & App. No. 22000001422– 800 MW) – Barmer-III PS 400 kV D/c line (suitable to carry minimum 900 MW per circuit at nominal voltage) along with associated bays at generation end.

C. Transmission system for Connectivity under GNA: As per **Annexure-I**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system										

Connectivity at Bhadla Complex:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
3.	2200001417	Sprng Green Energy 4 Private Limited	Bikaner distt., Rajasthan	29.10.2024	Generator (Solar)	Land BG Route	400	30.06.2029	Bhadla-IV PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Sprng Green Energy 4 Private Limited has applied for connectivity for 4000 MW at Bhadla-IV PS. However, as informed in the 36th CMETS NR meeting with grant of 5825 MW connectivity at Bhadla-IV PS, Bhadla-IV PS is closed for grant of additional connectivity except enhancements (upto 175 MW). Accordingly, it was proposed to grant connectivity at Bhadla-V PS through 220 kV S/c line.

Applicant was asked to confirm the scope of 220 kV line bay at Bhadla-V PS end & DTL tower configuration. M/s Sprng informed that 220 kV bay at Bhadla-V end shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. Same was noted. Further, M/s Sprng was asked to submit necessary undertaking for S/c line on D/c tower configuration.

It was mentioned that the transmission system for evacuation of power from Bhadla-V PS is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Sprng Green Energy 4 Private Limited has requested Start Date of Connectivity under GNA from 30.06.2029. However, considering the same and expected commissioning date of transmission system (i.e. 30.06.2031), the start date of Connectivity under GNA shall be 30.06.2031(interim). Transmission system for connectivity of M/s Sprng Green Energy 4 Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

(i). Sprng Green Energy 4 Private Limited RE Power Project – Bhadla-V PS 220 kV S/c line on D/c tower (suitable to carry minimum 400 MW at nominal voltage) along with associated bay at generation end.

C. <u>Transmission system for Connectivity under GNA:</u> As per **Annexure-II**

Start Date of Connectivity under GNA: 30.06.2031(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
4.	2200001420	Azure Power Fifty One Private Limited	Jodhpur distt., Rajasthan	30.10.2024	Generator (Solar)	Land Route	600	31.12.2029	Bhadla-IV PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Azure Power Fifty One Private Limited has applied for connectivity for 600 MW at Bhadla-IV PS. However, as informed in the 36th CMETS NR meeting, with grant of 5825 MW connectivity at Bhadla-IV PS, Bhadla-IV PS is closed for grant of additional connectivity except enhancements (upto 175 MW). Accordingly, it was proposed to grant connectivity at Bhadla-V PS through 400 kV S/c line.

Applicant was asked to confirm the scope of 400 kV line bay at Bhadla-V PS end & DTL tower configuration. M/s Azure Power in the meeting requested CTUIL to grant connectivity of 600 MW at 220 kV level. CTUIL informed that considering the connectivity quantum(600 MW), connectivity at 400 kV is optimal compared to 220 kV level. Therefore, considering the optimal utlisation of ISTS system & RoW it is considered to grant connectivity at 400 kV.

Subsequently, M/s Azure vide letter dated informed that they have land in procession for 300 MW in vicinity of Bhadla Complex & further expansion of the land parcel is not feasible due to existing projects in the surrounding area. Therefore, they had planned to procure land for additional 300 MW capacity after finalization of location of Pooling station. In view of the above, M/s Azure again requested to grant connectivity at 220 kV.

Considering the above aspects, it was decided to take up the above application by M/s Azure again for discussion in the next CMETS NR meeting.

C2. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Nov'24

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200001440	Pokhran Wind Energy Private Limited	Goonga, Jaisalmer, Rajasthan	04-11-2024	Generator (Wind)	Land BG Route	99	31-01-2027	Fatehgarh PS
2.	2200001441	Hexa Climate Solutions Private Limited	Khichya, Bikaner, Rajasthan	06-11-2024	Generator (Solar)	Land BG Route	50	31-03-2026	Bikaner PS
3.	2200001446	Aditya Birla Renewables Limited	Lakha, Barmer, Rajasthan	07-11-2024	Generator (Hybrid) (Solar: 25 MW, Wind: 25 MW)	Land BG Route	50	15-06-2027	Barmer-II PS
4.	2200001449	TEQ Green Power XIII Private Limited	Jhanphli Kalan and adjoining villages, Barmer, Rajasthan	09-11-2024	Generator (Solar)	REMCL LOA	50	30-11-2026	Barmer PS
5.	2200001464	Avaada Energy Private Limited	Bikampur, Bikaner, Rajasthan	12-11-2024	Generator (Solar)	NHPC LOA	1000	30-06-2027	Bhadla IV PS
6.	2200001503	Renew Solar Power Private Limited	Bhadkha, Barmer, Rajasthan	22-11-2024	Generator (Solar)	NTPC LOA	300	31-01-2027	Barmer PS
7.	2200001523	Tata Power Renewable Energy Limited	Ramgarh, Jaisalmer, Rajasthan	28-11-2024	Generator (Wind)	Land BG Route	300	31-03-2027	Ramgarh PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
1.	2200001440	Pokhran Wind Energy Private Limited	Goonga, Jaisalmer, Rajasthan	04-11-2024	Generator (Wind)	Land BG Route	99	31-01-2027	Fatehgarh PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Pokhran Wind Energy Private Limited has applied for connectivity for 99 MW at Fatehgarh PS. However, there is no margin available at Fatehgarh PS. Further, margin is available only at Barmer-III PS in Fatehgarh- Barmer Complex. Accordingly, it was proposed to grant connectivity at 220 kV Barmer-III PS in sharing with M/s Foxtrot Solar Renewable Energy Pvt. Ltd. which is already granted connectivity of 254 MW(100+80+74) at Barmer-III PS.

M/s Pokhran Wind Energy in the meeting informed that they have plan to apply for additional connectivity in the same pooling station and therefore requested CTUIL to grant connectivity through a separate 220 kV bay. Regarding this, CTUIL informed that more than 4000 MW of connectivity is already granted/agreed for grant in Barmer-III PS. With the applications considered in the present meeting total connectivity at Barmer-III PS is close to 6000 MW. As Barmer-III PS is being planned for

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

6000 MW capacity only, the connectivity capacity at Barmer-III PS is almost fully exhausted. Therefore no additional connectivity can be granted beyond 6000 MW at Barmer-III PS. Therefore, considering the optimal utilization of bays allocated at Barmer-III PS as well as the connectivity quantum of M/s Pokhran Wind(99 MW) , it was proposed to grant connectivity at 220 kV Barmer-III PS in sharing with M/s Foxtrot Solar Renewable Energy Pvt. Ltd. M/s Pokhran Wind agreed for the same. M/s Pokhran Wind need to submit sharing agreement with M/s Foxtrot Solar for sharing of bay & DTL. M/s Pokhran Wind noted the same.

It was mentioned that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Pokhran Wind Energy Private Limited has requested Start Date of Connectivity under GNA from 31.01.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim).

Accordingly, it was agreed to grant connectivity of 99 MW to M/s Pokhran Wind Energy Private Limited at Barmer-III PS in sharing with M/s Foxtrot Solar at 220 kV. Details of transmission system for connectivity of M/s Pokhran Wind Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. Associated Transmission System (ATS): NIL

B. Transmission System under applicant scope

(i). Through sharing of dedicated transmission of M/s Foxtrot Solar Renewable Energy Pvt. Ltd. RE Power Projects (App. No. 2200001057-100 MW, 2200001088 – 80 MW, 22000001252 – 74 MW) – Barmer-III PS 220 kV S/c line on D/c tower(Suitable to carry minimum 353 MW at nominal voltage)

(ii). Infrastructure required for sharing dedicated transmission system - under the scope of M/s Pokhran Wind Energy Private Limited

C. Transmission system for Connectivity under GNA: As per **Annexure-I**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
2.	2200001441	Hexa Climate Solutions Private Limited	Khichya, Bikaner, Rajasthan	06-11-2024	Generator (Solar)	Land BG Route	50	31-03-2026	Bikaner PS	Not Applicable

It was informed that M/s Hexa Climate Solutions Private Limited has applied for connectivity of 50 MW at Bikaner PS. At Bikaner PS, margin of 50 MW is available only at 400 kV level in sharing with M/s Renew, M/s Azure or M/s Ayana. Considering the coordinates provided in the application, it was observed that project location is very close to Ayana Renewable. Further, the location is within 10 km from the other two 400 kV grantees at Bikaner PS(Renew & Azure). Accordingly, connectivity can be granted at Bikaner PS at 400 kV level through sharing of dedicated transmission system of either of Renew/Azure/Ayana which is to be confirmed by the applicant. Therefore, applicant was asked to confirm the name of entity whose DTL they shall be sharing for connectivity of 50 MW.

M/s Hexa Climate informed that they are in discussion with M/s Renew Solar & M/s AK Renewable infra(entity granted in sharing with M/s Renew) for sharing of DTL Accordingly, it was proposed to grant connectivity at 400 kV in sharing with M/s Renew Solar. CTUIL informed that earlier also M/s Hexa had files a similar application for 50 MW connectivity at Bikaner PS which was closed due to non submission of Conn BGs.

CTUIL informed that to avoid issues in sharing of dedicated transmission system connectivity in such cases, it is advised to have discussions with existing connectivity grantees and enter into sharing agreement before applying for connectivity.

It was also informed that M/s Hexa Climate Solutions Private Limited has requested Start Date of Connectivity under GNA from 31.03.2026. Considering the same and expected commissioning date of transmission system (i.e. 31.03.2027), the start date of Connectivity under GNA shall be 31.03.2027 (interim). M/s Hexa Climate noted the same.

Accordingly, in the meeting it was agreed to grant connectivity of 50 MW to M/s Hexa Climate at Bikaner PS in sharing with M/s Renew at 400 kV.

Subsequently, M/s Hexa Climate vide mail dated 30.04.2025 informed their decision to withdraw the above connectivity application. Accordingly, it was decided to close the above application as per GNA Regulations.

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
3.	2200001446	Aditya Birla Renewables Limited	Lakha, Barmer, Rajasthan	07-11-2024	Generator (Hybrid) (Solar: 25 MW, Wind: 25 MW)	Land BG Route	50	15-06-2027	Barmer-II PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 2 lakh/MW

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

It was informed that M/s Aditya Birla Renewables Limited has applied for connectivity for 50 MW at Barmer-II PS. However, there is no margin available at Barmer-II PS. Further, 150 MW of connectivity of M/s Unique Hytech was agreed to be granted in sharing with M/s TEQ Green (208.5MW) at 220 kV level of Barmer-III PS in 36th CMETS NR meeting. Subsequently, M/s Unique Hytech applied for additional connectivity of 1000 MW (App. No. 2200001622) at Barmer-III PS in the month of Dec'24 which was also to be discussed in present meeting. Considering above, the proposal for revision of connectivity to be granted to M/s Unique Hytech(150 MW at 220 kV level) was discussed in same meeting for sharing its connectivity with new application of M/s Unique Hytech (1000MW) at 400kV level of Barmer-III PS.

M/s Unique Hytech in the meeting informed that they agree to shift of their earlier granted connectivity of 150 MW to 400 kV in Barmer-III PS in sharing with their 1000 MW application which is proposed to be granted at 400 kV Barmer-III PS in the same meeting.

In view of the above, 150 MW margin allocated to Unique Hytech at 220 kV bay in sharing is now vacated. Therefore, 50 MW can be allocated to M/s Aditya Birla at the same 220 kV bay in sharing with M/s TEQ Green. Accordingly, it was proposed to grant 50 MW connectivity to M/s Aditya Birla at Barmer-III PS in sharing with M/s TEQ Green Power IX Pvt. Ltd (App. No. 2200001300 – 208.5 MW) at 220 kV Barmer-III PS. M/s Aditya Birla agreed for the same. M/s Aditya Birla need to submit sharing agreement with M/s TEQ Green for sharing of bay & DTL. M/s Aditya Birla noted the same.

It was mentioned that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Aditya Birla Renewables Limited has requested Start Date of Connectivity under GNA from 15.06.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim). M/s Aditya Birla Renewables Limited noted the same.

Accordingly, it was agreed to grant connectivity of 50 MW to M/s Aditya Birla Renewables Limited at Barmer-III PS in sharing with M/s TEQ Green Power IX Pvt. Ltd at 220 kV. Details of transmission system for connectivity of M/s Aditya Birla Renewables Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(i). Through sharing of dedicated transmission of M/s TEQ Green Power IX Pvt. Ltd. RE Power Project (App. No. 2200001300 – 208.5 MW) – Barmer-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage)

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

(ii). Infrastructure required for sharing dedicated transmission system - under the scope of M/s Aditya Birla Renewables Limited

C. **Transmission system for Connectivity under GNA:** As per **Annexure-I**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
4.	2200001449	TEQ Green Power XIII Private Limited	Jhanphli Kalan and adjoining villages, Barmer, Rajasthan	09-11-2024	Generator (Solar)	REMCL LOA	50	30-11-2026	Barmer PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Nil(Bay in sharing) • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s TEQ Green Power XIII Private Limited has applied for connectivity for 50 MW at Barmer PS. However, there is no margin available at Barmer-I PS & Barmer-II PS. Further, another entity of same group company M/s TEQ Green Power IX Pvt. Ltd. (App. No. 2200001300) is already granted connectivity of 208.5 MW at 220 kV Barmer-III PS through S/c line. Further, 150 MW of connectivity which was previously agreed to be granted to M/s Unique Hytech in sharing with M/s TEQ Green in 36th CMETS NR meeting is now revised for grant at 400 kV along with other application of Unique Hytech(1000 MW). Accordingly, it was proposed to grant 50 MW to M/s TEQ Green in sharing with earlier granted 220 kV bay with App. No. 2200001300(208.5 MW).

M/s TEQ Green Power XIII in the meeting informed that they wish to withdraw the above application. The same was confirmed by M/s TEQ Green vide mail dated 01.04.2025. Accordingly, it was decided to close the above application as per GNA Regulations 2022.

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
5.	2200001464	Avaada Energy Private Limited	Bikampur, Bikaner, Rajasthan	12-11-2024	Generator (Solar)	NHPC LOA	1000	30-06-2027	Bhadla IV PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Avaada Energy Private Limited has applied for connectivity for 1000 MW at Bhadla-IV PS. However, as informed in the 36th CMETS NR meeting with grant of 5825 MW connectivity at Bhadla-IV PS, Bhadla-IV PS is closed for grant of additional connectivity except enhancements (upto 175 MW).

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

Accordingly, it was proposed to grant connectivity at Bhadla-V PS through 400 kV S/c line. M/s Avaada agreed for the same.

Applicant was asked to confirm the scope of 400 kV line bay at Bhadla-V PS end & tower configuration. M/s Avaada informed that 400 kV Bays at Bhadla-V PS shall be under ISTS

It was mentioned that the transmission system for evacuation of power from Bhadla-V PS is presently under evolution. Upon evolution, the scheme shall be discussed in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Avaada Energy Private Limited has requested Start Date of Connectivity under GNA from 30.06.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.06.2031), the start date of Connectivity under GNA shall be 30.06.2031(interim). M/s Avaada Energy noted the same.

Accordingly, it was agreed to grant connectivity of 1000 MW to M/s Avaada Energy Private Limited at Bhadla-V PS at 400 kV. Details of transmission system for connectivity of M/s Aditya Birla Renewables Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(i). Avaada Energy Private Limited RE Power Project – Bhadla-V PS 400 kV S/c line on D/c tower (suitable to carry minimum 1000 MW at nominal voltage) along with associated bay at generation end

C. **Transmission system for Connectivity under GNA:** As per **Annexure-II**

Start Date of Connectivity under GNA: 30.06.2031(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
6.	2200001503	Renew Solar Power Private Limited	Bhadkha, Barmer, Rajasthan	22-11-2024	Generator (Solar)	NTPC LOA	300	31-01-2027	Barmer PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: As per bay scope • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Renew Solar Power Private Limited has applied for connectivity for 300 MW at Barmer-I PS. However, there is no margin available at Barmer-I PS & Barmer-II PS. Accordingly, it was proposed to grant connectivity at Barmer-III PS. It is to mention that M/s Renew Solar is already granted connectivity of 200 MW(App. No. 2200001026) & 250 MW(App. No. 2200001188) at 220 kV Barmer-III PS. Considering the present application of 300 MW, total connectivity shall become 750 MW. Therefore, considering optimal utilization of RoW, it was proposed to grant connectivity to Renew at 400 kV level of Barmer-III PS through 1 no. of bay for the combined capacity of 750 MW(200MW+250MW+300MW). M/s Renew agreed for the same.

M/s Renew Solar was asked confirm the scope of 400 kV line bay at Barmer-III PS end & tower configuration. Further, M/s Renew was also asked to confirm the lead entity among these for submission of Conn BG-2(Rs. 6 Cr.) for 400 kV bay (if agreed under ISTS). M/s Renew informed that they shall update regarding the same after the meeting.

Subsequently, M/s Renew vide mail & letter dated 08.04.2025 informed that the tower configuration for 400 kV line shall be S/c line on D/c towers & 1 no. 400kV line bay at Barmer-III PS end shall be under ISTS. Further, lead entity for submission of Conn-BG2 amounting to Rs. 6.0 Crores, towards the 400 kV bay at Barmer-III PS, shall be ReNew Solar Power Private Limited, bearing Application No. 2200001026 (200 MW). The same was noted.

Further, it was also informed in the meeting that earlier submitted Conn BG-2 for 220 kV bays associated with application No. 2200001026 & 2200001188 shall be released after submission of revised Conn BG-2 of Rs. 6 Cr. for lead entity. The revised intimations for application No. 2200001026 & 2200001188 shall be issued accordingly. M/s Renew noted the same.

It was mentioned that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Renew Solar Power Private Limited has requested Start Date of Connectivity under GNA from 31.01.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim). M/s Renew Solar Power Private Limited noted the same.

Accordingly, it was agreed to grant connectivity of 300 MW to M/s Renew Solar Power Private Limited at Barmer-III PS at 400 kV along with other applications of M/s Renew of 200 MW(App. No. 2200001026) & 250 MW(App. No. 2200001188). Details of transmission system for connectivity of M/s Renew Solar Power Private Limited under GNA is as below:

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(i). Common Pooling Station of Renew Solar Power Private Limited RE Power Projects (App. No. 2200001026-200 MW, App. No 2200001188-250 MW & App. No. 2200001503-300 MW) – Barmer-**III** PS 400 kV S/c line on D/c tower (suitable to carry minimum 900 MW at nominal voltage) along with associated bay at generation end

C. **Transmission system for Connectivity under GNA:** As per **Annexure-I**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
7.	2200001523	Tata Power Renewable Energy Limited	Ramgarh, Jaisalmer, Rajasthan	28-11-2024	Generator (Wind)	Land BG Route	300	31-03-2027	Ramgarh PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Tata Power Renewable Energy Limited has applied for connectivity for 300 MW at Ramgarh PS. However, there is no margin available at Ramgarh PS. Accordingly, it was proposed to grant connectivity at Ramgarh-II PS through 220 kV S/c line.

Applicant was asked to confirm the scope of 220 kV line bay at Ramgarh-II PS end & DTL tower configuration. Regarding this, M/s Tata Power informed that the scope of 220 kV line bay at Ramgarh-II PS shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Ramgarh-II PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Tata Power Renewable Energy Limited has requested Start Date of Connectivity under GNA from 31.03.2027. However, considering the same and expected commissioning date of transmission system (i.e. 31.03.2030), the start date of Connectivity under GNA shall be 31.03.2030 (interim).

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

Accordingly, it was agreed to grant connectivity of 300 MW to M/s Tata Power Renewable Energy Limited at Ramgarh-II PS at 220 kV . Details of transmission system for connectivity of M/s Tata Power Renewable Energy Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

(ii). Tata Power Renewable Energy Limited RE Power Project – Ramgarh-II PS 220 kV S/c line on D/c tower (suitable to carry minimum 300 MW at nominal voltage) along with associated bay at generation end

C. **Transmission system for Connectivity under GNA:** As per **Annexure-III**

Start Date of Connectivity under GNA: 31.03.2030(Interim). Final date shall be confirmed upon award of the system

B3. Applications for Connectivity to ISTS under CERC (Connectivity and General Network Access to the inter-State Transmission System) Regulations, 2022 received in Dec'24

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)
1.	2200001557	Apraava Energy Private Limited	Jhanphli Kalan, Barmer, Rajasthan	10-12-2024	Generator (Solar)	Land BG Route	300	31-12-2027	Fategarh Barmer ISTS Complex
2.	2200001622	Unique Hytech Renewable Energy Private Limited	Moseri, Karni Nagar, Jhanakali, Barmer, Rajasthan	28-12-2024	Generator (Hybrid) (Solar: 400 MW, Wind: 600 MW))	Land BG Route	1000	31-07-2030	Barmer – III PS

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
1.	2200001557	Apraava Energy Private Limited	Jhanphli Kalan, Barmer, Rajasthan	10-12-2024	Generator (Solar)	Land BG Route	300	31-12-2027	Fategarh Barmer ISTS Complex	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 3 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Apraava Energy Private Limited has applied for connectivity for 300 MW at Fategarh Barmer ISTS Complex. Accordingly, it was proposed to grant connectivity at Barmer-III PS through 220 kV S/c line.

Applicant was asked to confirm the scope of 220 kV line bay at Barmer-III PS end & tower configuration. Regarding this, M/s Apraava Energy informed that the scope of 220 kV line bay at Barmer-III PS shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Apraava Energy Private Limited has requested Start Date of Connectivity under GNA from 31.12.2027. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim).

Accordingly, it was agreed to grant connectivity of 300 MW to M/s Apraava Energy Private Limited at Barmer-III PS at 220 kV. Details of transmission system for connectivity of M/s Apraava Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

   (i). Apraava Energy Private Limited RE Power Project – Barmer-III PS 220 kV S/c line on D/c tower (suitable to carry minimum 300 MW at nominal voltage)

C. **Transmission system for Connectivity under GNA:** As per **Annexure-I**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement
2.	2200001622	Unique Hytech Renewable Energy Private Limited	Moseri, Karni Nagar, Jhanakali, Barmer, Rajasthan	28-12-2024	Generator (Hybrid) (Solar: 400 MW, Wind: 600 MW))	Land BG Route	1000	31-07-2030	Barmer – III PS	• Conn BG1: Rs. 50 Lakh • Conn BG2: Rs. 6 Cr. • Conn BG3: Rs. 2 lakh/MW

It was informed that M/s Unique Hytech Renewable Energy Private Limited has applied for connectivity for 1000 MW at Barmer-III PS. Further, as mentioned earlier, connectivity of 150 MW to M/s Unique Hytech already agreed for grant in 36th CMETS NR meeting held on 15.01.2025 at 220 kV Barmer-III PS in sharing with M/s TEQ Green Power IX was reviewed and considered for grant at 400 kV along with the present application through 400 kV S/c line for the combined capacity of 1150 MW(1000MW+150MW) . Accordingly, it was proposed to grant connectivity of 1150 MW(1000MW+150MW)at Barmer-III PS through 400 kV S/c line. M/s Unique Hytech agreed for the same.

Applicant was asked to confirm the scope of 400 kV line bay at Barmer-III PS end & tower configuration. Regarding this, M/s Unique Hytech Renewable informed that the scope of 400 kV line bay at Barmer-III PS shall be under ISTS & DTL tower configuration shall be S/c line on D/c tower. The same was noted.

It was mentioned that the transmission system for evacuation of power from Barmer-III PS is presently under discussion. The scheme is currently tentative, which shall be finalized in a separate joint study meeting among CEA, CTUIL, Grid India & Stakeholders followed by the subsequent CMETS meeting. The detailed transmission system shall be informed upon finalization and approval of the scheme.

It was also informed that M/s Unique Hytech Renewable Energy Private Limited has requested Start Date of Connectivity under GNA from 31.07.2030. However, considering the same and expected commissioning date of transmission system (i.e. 30.09.2030), the start date of Connectivity under GNA shall be 30.09.2030 (interim).

Accordingly, it was agreed to grant connectivity of 1000 MW to M/s Unique Hytech Renewable Energy Private Limited at Barmer-III PS at 400 kV. Further, 150 MW connectivity already granted to M/s Unique Hytech (App. No. 2200001407) shall be granted in sharing with above application of 1000 MW. Details of transmission system for connectivity of M/s Unique Hytech Renewable Energy Private Limited under GNA is as below:

Details of Transmission system for Connectivity under GNA:

A. **Associated Transmission System (ATS): NIL**

B. **Transmission System under applicant scope**

Sl. No.	Application ID	Applicant	Project Location	Submission Date	Nature of Applicant	Criterion for applying	Connectivity Quantum (MW)	Start Date of Connectivity (As per Application)	Connectivity location (As per Application)	Conn BGs requirement

(i). Common Pooling Station of M/s Unique Hytech Renewable Energy Private Limited REPower Projects (App. No. 2200001407 -150 MW & App. No. 2200001622 – 1000 MW) – Barmer-III PS 400 kV S/c line on D/c tower (suitable to carry minimum 1150 MW at nominal voltage)

C. <u>Transmission system for Connectivity under GNA:</u> As per **Annexure-I**

Start Date of Connectivity under GNA: 30.09.2030(Interim). Final date shall be confirmed upon award of the system.

It may be noted that with this grant of 1000 MW to M/s Unique Hytech, total connectivity granted at Barmer-III PS is 5961.5 MW. As Barmer-III PS was planned for 6000 MW, Barmer-III PS shall be closed for fresh connectivity grant except enhancements (upto 39.5 MW)

B4. Application for grant of GNA_{RE} received in the month of Dec'24

Sl No.	Application ID	Name of the Applicant	Submission Date	Nature of applicant	GNA_{RE} within Region (MW)	GNA_{RE} outside Region (MW)	Total GNA_{RE} Required (MW)	Start date of GNA_{RE}	End date of GNA_{RE}
1.	2200001564	Jubilant Ingrevia Limited	16-12-2024	Drawee entity connected to Intra State Transmission System	0	5.4	5.4	15-05-2025	31-03-2050

It was informed that M/s Jubilant Ingrevia Limited has applied for GNA_{RE} of 5.4 MW (Outside the region) as a drawee entity connected to Intra State Transmission System of UPPCL/PVVNL at 132kV Gajraula Substation.

It was mentioned that Uttar Pradesh has been granted GNA of 10339 MW as on date. UPPTCL has also applied for additional GNA of 799 MW under Reg. 19 of GNA Regulations 2022. Further, 4.5 MW of GNA_{RE} is also granted to intra state connected entities in Uttar Pradesh. Applicant has submitted NoC from UPPTCL. Considering above, there is sufficient margin available in the existing ISTS system for drawl of additional 5.4 MW from ISTS periphery of UPPTCL network. In view of the above, it was proposed to grant 5.4 MW of GNA_{RE} (outside the region) to Jubilant Ingrevia Limited.

As per application, M/s Jubilant Ingrevia had requested Start date of GNA_{RE} from 01.04.2025. However, in the meeting, M/s Jubilant Ingrevia requested to revise the start date from 01.05.2025. Subsequently, M/s Jubilient Ingrevia vide mail dated 24.04.2024 informed that they wish to revise their start date of GNA_{RE} from earlier mentioned 01.05.2025 to 15.05.2025. Accordingly, it was decided to grant GNA_{RE} of 5.4 MW(Outside region) to M/s Jubiliant Ingrevia Limited with Start date from 15.05.2025.

B5. Application for Additional Grant of GNA to STUs under Regulation 19:

Sl No.	Application ID	Name of the Applicant	Submission Date	Region	Nature of applicant	Additional GNA (MW)	GNA within Region (MW)	GNA outside Region (MW)	Start Date of GNA
1.	2200001286	U.P. Power Transmission Corporation Limited	26.09.2024	NR	State Transmission Utility (STU)	FY: 2025-26 – 699 FY: 2026-27 –100 FY: 2027-28 - 0	FY: 2025-26 –0 FY: 2026-27 –0 FY: 2027-28 - 0	FY: 2025-26 – 699 FY: 2026-27 –100 FY: 2027-28 - 0	699 MW: 01.04.2025 100 MW: 01.04.2026

Entity wise Segregation of GNA:

FY: 2025-26: UPPCL 600 MW, NPCL: 50 MW & Northern Railway:49 MW (Total: 699 MW)

FY: 2026-27: NPCL: 100 MW

It was stated that application for 799 MW GNA by UPPTCL(STU) received in the month of Sep'24 was discussed in the 35th CMETS NR meeting held on 29.10.2024. In the 35th CMETS NR meeting it was informed that, CTUIL is in the process of finalising GNA Study files for next three financial years (FY 2025-26, 2026-27 & 2027-28). In this regard, all STUs were requested to update their intrastate system in 2025-26, 2026-27 & 2027-28 timeframe files and share it with CTUIL. Subsequently, GNA study files were circulated to all STUs for further comments. No comments were received by CTUIL. Accordingly, studies were carried out for grant of additional 799 MW to UPPTCL.

To process the grant of GNA to UPPTCL, Peak drawl scenario of Uttar Pradesh in FY 2025-26 was considered for studies. From the studies, it was observed that with peak drawl scenario of 17000 MW of UP, 400/220 kV ICT (3x315 MVA) at Allahabad(PG) is N-1 non-compliant (320MW in N-1 contingency). Loading of Agra (PG) and Lucknow (PG) ICTs are on higher side but within N-1 limit. No other constraints were observed in UP in the GNA study file.

The present Available Transfer Capability (ATC) of Uttar Pradesh declared by Grid India is about 16700 MW. At present, total GNA granted to UP is about 10339 MW. As per TTC/ATC study results of Grid-India, limiting constraints in ATC is N-1 contingency of 400/220kV Obra, Allahabad(PG), Gorakhpur (UP), Agra(PG) and Lucknow (PG) ICTs. It may be noted that the 4th ICT at Allahabad(PG) and Agra (PG) is under implementation and is expected to be commissioned by 31.03.25 and 20.10.26 respectively whereas Lucknow (PG) ICT is being taken up for approval. Powergrid stated that Allahabad ICT will be commissioned by Jun'25 . UPPTCL stated that 765/400kV 1x1000MVA ICT (1st) at Obra S/s is already commissioned in Oct'24 and 2nd 1000MVA ICT will be commissioned by Jun'25. At Gorakhpur (UP) existing 240MVA ICT was replaced with 500MVA ICT which will resolve the issue of N-1 non compliance at Gorakhpur (UP).

Grid-India enquired to UPPTCL that whether proposed additional GNA of 699MW is distributed all over the UP region or concentrated at some drawl point. UPPTCL stated that proposed GNA quantum is distributed at multiple drawl points in Uttar Pradesh.

Further in Grid-India files, loading of critical ICTs in UP were also reviewed. Loadings are as under:

Substation	ICTs Loading (in MW)-Grid India files

	Jun'24 **(UP drawl through tie lines: 16.3GW)**	**Apr'24** **(UP drawl through tie lines: 11.7GW)**
400/220kV Allahabad (PG) (3x315MVA)	3x239 (N-1: 284/ICT)	3x223 (N-1: 266/ICT)
400/220kV Lucknow (PG) (2x500MVA)	2x310 (N-1: 438)	2x244 (N-1: 347)
400/220kV Agra (PG) (2x315MVA)	2x170 (N-1: 235)	2x135 (N-1: 186)

From the loadings, its is envisaged that there is appreciable relief in ICT loading of Agra, Allahabad and Lucknow ICTs with lesser UP drawl. Cosidering studies outcome, it is seen that there is sufficient margin available in existing system for grant of additional GNA of 699 MW to UP STU in 2025-26 timeframe.

It is to mention that, for FY25-26, out of the 699 MW applied by UPPTCL, 50 MW of GNA is allocated to Noida Power Company Limited(NPCL). NPCL was granted additional GNA of 100 MW in the 32nd CMETS NR meeting with Start Date from 30.07.2024 upto 31.03.2026.

In the 32nd CMETS NR meeting it was highlighted that, as per the Operational Feedback Report of Grid India, issue of critical loading of 400 kV Dadri-Gr Noida(UP) D/c line is reported. In the operational feedback, it is mentioned that though 400kV Dadri Th- Gr. Noida is a quad moose line and can carry power upto ~1750–1800 MW, it has been observed that after by-passing of 400kV Dadri-Maharanibagh-Ballabgarh at 400kV Maharanibagh, loading of 400 kV Dadri Th- Gr. Noida has further increased slightly & loading of upto 1500 MW is observed in the above line.

Regarding the loading of 400kV Dadri Th- Gr. Noida line, Grid India in 32nd CMETS NR meeting had informed that the loading of 400 kV Dadri – Greater Noida line has touched 1500 MW in 2024 but has some margins. However, considering the loading of other interconnecting lines of Dadri, it is not likely to breach N-1 limit in near future. As the GNA requested by M/s NPCL was only till Mar'26, the chances of any constraints arising till that time is very less. Further, as Noida Power was already scheduling the power through T-GNA, the same power will get shifted to GNA. Therefore, there won't be much change in the loading due to above grant of GNA. Considering the above aspects, it was agreed to grant GNA of 100 MW (Outside the region) to M/s NPCL.

As per the CTU studies loading of 400kV Dadri Th- Gr. Noida line is about 1500MW (in N-1 contingency) in higher UP drawl scenario (~17GW), however loadings will be relieved considerably (about 100-150MW) in lesser UP drawl scenario (~12GW), Currently, considering the additional 50 MW GNA being allocated to NPCL under the present application for FY2025-26, incremental loading of about 5-10 MW in 400 kV Dadri-Gr Noida(UP) D/c line may be observed. In present meeting, Grid-India stated that considering all facts as mentioned above, GNA may be granted to UPPTCL. No other comments were received in the meeting.

For FY26-27 UPPTCL has applied for 100 MW GNA with allocation to NPCL. 100 MW GNA granted to NPCL in 32nd CMETS NR meeting shall be expiring on 31.03.2026(End of FY 25-26). Therefore from 01.04.2026, the 100 MW additional GNA which is being granted with the present application shall replace the already granted GNA of 100 MW which is expiring on 31.03.2026. Therefore, there shall not be additional loading in the system due to this grant of GNA.

Considering above, it was agreed to grant GNA to UPPTCL(STU) with allocation as below:

i. FY 2025-26 - UPPCL 600 MW (Start Date: 01.04.2025)
ii. FY 2025-26 - NPCL 50 MW (Start Date: 01.04.2025)
iii. FY 2025-26 – Northern Railways 49 MW (Start Date: 01.04.2025)
iv. FY 2026-27 - NPCL 100 MW (Start Date: 01.04.2026)

C. Network Expansion Scheme

C1 Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) in Himachal Pradesh

Transmission system for evacuation of power from Shongtong Karcham HEP (450 MW) and Tidong HEP (150 MW) was recommended in the 11th meeting of NCT with RECPDCL as the BPC. Scheme has been notified vide Gazette notification dated 13.04.2023. Further the scheme was modified and approved in 17th NCT meeting held on 30.01.24. Details of scheme is as under :

A. Phase-I with Tidong HEP [Schedule: 01st July 2026]

- Establishment of 2x315 MVA (7x105 MVA 1-ph units including a spare unit) 400/220 kV GIS Pooling Station at Jhangi
- 400 kV Jhangi PS – Wangtoo (Quad) D/c line (Line capacity shall be 2500 MVA per circuit at Nominal voltage)

B. Phase-II with Shongtong HEP [Schedule: 31st July, 2026]

- LILO of one circuit of Jhangi PS - Wangtoo (HPPTCL) 400 kV D/c (Quad) line$ at generation switchyard of Shongtong HEP
- Panchkula- Point PW** 400 kV D/c line (Twin HTLS*) along with 80 MVAr switchable line reactor at Panchkula end on each circuit – 90 km
- Point PW** - Wangtoo (HPPTCL) 400 kV D/c line (Quad AL 59/Quad ACSR Moose/Quad AAAC) - 85 km
  (Total Route length-175 km)

***** Point PW : First point of 2000 m altitude of Panchkula-Wangtoo line from Panchkula end***
$ Line capacity shall be 2500 MVA per circuit at nominal voltage
**** with minimum capacity of 2100 MVA on each circuit at nominal voltage and Min.***

The above transmission scheme in under bidding with implementation timeframe of 30 months.

Subsequently, HPPTCL vide letter dated 16.01.2025 informed that HPPCL has intimated scheduled commissioning of Shongtong HEP in Nov'26 and transmission scheme (under TBCB) for evacuation of power from Shongtong HEP has not yet been awarded for execution (Copy of letter is attached in **Annexure D1**). As only 22 months are left for the proposed commissioning of Shongtong HEP, HPPTCL had proposed following interim arrangement for evacuation of power from Shongtong HEP:

- LILO of one circuit of Baspa-II — Karcham Wangtoo 400 kV line (Triple snowbird) at Shongtong HEP

Further, HPPTCL mentioned that as there is limited corridor in the narrow valley, the LILO portion may be constructed on Quad configuration under ISTS and the same would become part of the final transmission scheme. In the letter HPPTCL requested that the proposed Interim arrangement may kindly be got evaluated on priority and decision to construct the interim arrangement be taken at the earliest to avoid any generation loss

To deliberate on interim arrangement for evacuation of transmission system for evacuation of power from Shongtong Karcham HEP (450 MW), a meeting was held in CEA on 30.01.2025 **.** In the meeting, CEA stated that the similar interim arrangement was earlier agreed in a meeting held on 03.01.20. CEA opined that modalities and requirement of proposed interim arrangement could be ascertained after getting the firm commission timeframe of the Shongtong HEP and timeframe of transmission scheme that is under bidding. BPC (RECPDCL) informed that the Bid Evacuation Committee (BEC) had decided to go for re-bidding of the above transmission scheme, however, final view on the same is yet to be taken.

HPPCL informed that the works for Shongtong HEP are under full swing and they are trying best to achieve the commissioning of hydro project by November, 2026. In the meeting, JSW raised the issue of under rated XLPE cable (630 sqmm) at K'Wangtoo end of Baspa-II — K'Wangtoo 400 kV D/c line which limits the capacity of this Triple snowbird line. Upon query regarding the change of under rated cable, JSW stated that the cable delivery would take at least couple of years and long shutdown would also be required for the same. Further, JSW expressed concerns about congestion in the existing corridor due to the interim arrangement and opined for realignment of SPS setting in the complex.

HPPTCL stated that the interim arrangement could be established under ISTS and Shongtong HEP would shutdown one unit in case of outage in one ckt of Baspa-II — K'Wangtoo 400 kV D/c line. Further, the SPS scheme for the entire complex should be revised considering Shongtong HEP. HPPTCL highlighted the issue of bottlenecks in the existing system such as under rated cable, under rated bay equipment's etc. that limits the capacity of the corridor and suggested for measures for removal of those bottlenecks.

CTUIL in the meeting informed that from the load flow studies, it is envisaged that considering Shongtong generation with interim arrangement (LILO of one ckt of Baspa-II — Karcham Wangtoo at Shongtong HEP), loadings are in order; however, loading of 400 kV Nathpa Jhakri — Gumma — Panchkula D/c line (Triple snowbird) under N-1 contingency is about 950 MW. Considering huge quantum of hydro generation in above complex after additional generation of Shongtong HEP (450 MW) and Tidong HEP (150 MW), it may be prudent to plan additional corridor beyond Wangtoo (Wangtoo - Panchukula) to maintain system reliability & security. Also, SPS requirement and setting in above complex with proposed interim arrangement may be reviewed.

From the studies considering the interim arrangement, in the contingency of 400 kV Baspa - Shongtong circuit, loading would be about 825 MW (considering 10% overloading in peak hydro season in N-1 contingency). Therefore, due to the under rated cable (700 MW capacity) in existing 400 kV Baspa - K Wangtoo D/c line section, SPS arrangement on Baspa-Shongtong generation complex would also be required with above interim arrangement at the event of contingency. Also, JPTL may assess the possibility to upgrade the power rating of above cables.

NRLDC in the meeting stated that in case Shongtong and Tidong generation is to be evacuated before commissioning of 400kV Wangtoo — Panchkula D/c line, switchgear replacement for Jhakri — Panchkula section and Rampur — Nalagarh section may be carried out as it may lead to SPS requirement under N-1 contingency. Modification of SPS under different contingencies needs to be discussed with all stakeholders at OCC/NRPC level after revised simulation studies and also depending on switchgear upgradation work at POWERGRID hydro stations.

Shongtong HEP should confirm whether they will be able to receive SPS signal from Nathpa Jhakri/Rampur/Gumma and trip their units in case of N-1-1 contingency of lines, as the comprehensive SPS logic of the whole complex would require tripping of Shongtong generation. For replacement of terminal equipment, prolonged shutdown may be required. Shutdown requirement for terminal equipment replacement also to be discussed beforehand to take judicious call as most substations in the complex are GIS substations.
After further deliberations, following was concluded in the meeting :

- HPPTCL is requested to take the matter in the coming meeting of NRPC for consultation with all the stakeholders.
- Decision on the interim arrangement would be taken based on the timeline of commissioning of the planned transmission scheme and the timeline of commissioning of Shongtong Karcham HEP.

Subsequently, the matter was deliberated in the 228th OCC meeting of NRPC held on 14.02.25. In the meeting NRLDC mentioned the following :

- In case Shongtong and Tidong generation is to be evacuated before commissioning of 400kV Wangtoo – Panchkula D/c line, switchgear replacement for Jhakri – Panchkula section and Rampur – Nallagarh section may be carried out as it may lead to SPS requirement under N-1 contingency.
- Modification of SPS under different contingencies needs to be discussed with all stakeholders at OCC/NRPC level after revised simulation studies and also depending on switchgear upgradation work at POWERGRID/Hydro stations.
- Shongtong HEP should confirm whether they will be able to receive SPS signal from Nathpa Jhakri/Rampur/Gumma and trip their units in case of N-1-1 contingency of lines, as the comprehensive SPS logic of the whole complex would require tripping of Shongtong generation.
- For replacement of terminal equipment, prolonged shutdown may be required. Shutdown requirement for terminal equipment replacement also to be discussed beforehand to take judicious call as most substations in the complex are GIS substations

In the meeting it was concluded that interim arrangement i.e. LILO of one circuit of the Baspa-II – Karcham Wangtoo 400 kV line (Triple Snowbird) at Shongtong HEP must first be discussed in CMETS meeting and thereafter it may be taken up in the upcoming NRPC meeting as power of Shontong HEP would be evacuated through 400KV Karcham-Wangtoo-KalaAmb-Abdullapur section which has been implemented under ISTS. Further, a team of hydro experts from CEA may visit the site to assess the progress of the ongoing work and likely commissioning of the Shontong HEP (450MW) project. Based on their assessment, a decision could be made on whether to proceed with the proposed interim arrangement or as per originally planned evacuation system.

Subsequently, HPPTCL vide letter dated 22.02.25 to CEA submitted that there is only a left out period of 20 months for the intimated commissioning of Shongtong HEP i.e. November, 2026 and implementation of transmission system for evacuation of Shongtong HEP is yet to be awarded by the concerned authorities and it is now anticipated that the transmission system shall not be available for evacuation by the day the project is commissioned even if it is three to four months beyond November, 2026. Keeping this in view, HP has suggested the implementation of interim arrangement for evacuation which too may take around 24 months for implementation. As conveyed by NRPC vide his letter dated 19.02.2025, the decision to be taken for taken up the proposed interim arrangement based upon the outcome of the assessment to be done by the team of Hydro experts from CEA after their field visit to the site of Shongtong HEP. CEA may update outcome of the assessment to be done by the team of Hydro experts from CEA after their field visit to the site of Shongtong HEP

HPPTCL further requested that the decision to implement the proposed interim arrangement on RTM mode be got conveyed immediately irrespective of the outcome of the assessment of the committee of the hydro experts as the interim arrangement shall be part of the final transmission system and will cover slippage if any, in implementation of the transmission system to avoid generation loss as well as to give comfort to the developer.

Further the requirement of interim arrangement was deliberated in 28th NCT meeting held on 05.03.25 wherein it was decided that the interim arrangement (LILO (with quad conductor) of one ckt. of 400 kV Karcham Wangtoo – Baspa D/c line at generation switchyard of Shongtong HEP) to be implemented by the State (HPPTCL). The cost for the common portion which can be utilized later in final arrangement will be communicated by HPPTCL to CEA/CTUIL within 10 days. The bidding to continue as per original scope, however, the cost for the common portion should be included in the RfP. HPPTCL would recover the same from the successful bidder.

Based on OCC and NCT deliberations, technical approval for interim arrangement is taken up in this meeting highlighting issues raised in earlier meetings by NRLDC/CTU for which following information was sought :

- POWERGRID may confirm the rating of existing 400kV bays at Nalagarh end of 400kV Rampur – Nallagarh D/c line and Panchkula S/s end of N'Jhakri - Panchkula D/c line (LILO at Gumma) along with space confirmation for bay upgradation at Nalagarh, Rampur, N'Jhakri and Panchkula S/s – In the meeting, PowerGrid stated that both Nalagarh and Panchkula are AIS S/s having switchgear rating of 2kA bay and it is feasible to upgrade it to 3.15kA. Powergrid requested that in case of bay upgradation works taken up, existing bay equipment may be treated as regional spare. CTU stated that treatment of old bay equipment's along with approval of upgradation works in ADD-CAP shall be discussed in OCC/NRPC meeting.

- HPPTCL may also confirm the rating of existing 400kV bays at Gumma S/s for 400kV N'Jhakri – Gumma-Panchkula D/c line along with space confirmation for bay upgradation at Gumma S/s – HPPTCL stated that both Gumma and Wangtoo S/s have higher rating and need not require upgradation.
- SJVN may confirm the rating of existing 400kV bays at Rampur end of 400kV Rampur – Nallagarh D/c line and N'Jhakri S/s end of N'Jhakri - Panchkula D/c line (LILO at Gumma) along with space confirmation for bay upgradation at Rampur, and N'Jhakri – In the meeting, SJVN stated that both N'Jhakri and Rampur are old GIS substation and having switchgear rating of 2kA (breaker capacity) which is sufficient to carry 950MW in N-1 contingency, however, upgradation of old GIS equipment is very risky and time consuming as well as having shutdown requirement as per their past experience.

CTU requested POWERGRID, SJVN and HPPTCL to provide the information through mail regarding the existing bay equipment rating (CT/isolator/CB) in the desired format circulated by CTU before taken up the agenda in NRPC meeting.

NRLDC stated that SPS in above hydro complex is designed considering above limitation of bay switchgear equipment's at various substations. NRLDC stated that for SPS implementation with Shongtong HEP, HPPTCL/Shongtong may provide information that how they will receive signal upto their generation switchyard.

SJVN stated that SPS signal is already extended to N. Jhakri to K Wangtoo S/s. CTU stated that as per their information, Baspa-K'Wangtoo has optical fibre, however confirmation may be taken from POWERGRID/JPL. CTU stated that OPGW may be considered by HPPTCL in interim arrangement i.e. LILO of one ckt of 400kV Karcham Wangtoo- Baspa D/c line at Shongtong HEP. HPPTCL stated that they are considering OPGW in interim portion and any additional requirement will be incorporated as part of interim arrangement for SPS implementation. CTU stated that modalities of communication signal up to shongtong HEP may be discussed in next #NRPC TeST meeting. HPPTCL agreed for the same to take up the agenda in NRPC TeST meeting after CMETS-NR minutes. Grid-India further reiterated that modification of SPS in above complex as well as local SPS in 400kV Baspa- K'wangtoo line under different contingencies needs to be discussed with all stakeholders at OCC/NRPC level after revised simulation studies.

Further , it was stated that from the load flow studies that considering Shongtong generation with interim arrangement (LILO of one ckt of Baspa-II — Karcham Wangtoo at Shongtong HEP), loadings are in order. Loading of 400 kV Nathpa Jhakri — Gumma — Panchkula D/c line (Triple snowbird) under N-1 contingency is about 950 MW for which SPS requirement may be reviewed by NRLDC. CEA and CTU requested to HPPTCL to expedite the submission of costing to CEA/CTU as informed in 28th NCT meeting considering same conductor configuration of interim arrangement system inline with final arrangement (Line capacity of LILO of one ckt of 400kV Karcham Wangtoo- Baspa D/c line Shongtong HEP shall be 2500 MVA per circuit at nominal voltage).

HPPTCL agreed for the same and stated that they are working on cost and will provide the same in next one week. HPPTCL also stated that in 78th NRPC meeting held on 16.03.25 they have requested that Construction of S/c LILO of 400 kV D/C Baspa - Karcham Wangtoo line at Shongtong HEP under Regulated Tariff Mechanism which will later become part of final plan approved by CTUIL. Same may be noted in this meeting also.

CTU stated that the decision for implementation of interim arrangement by HPPTCL is already taken in 28th NCT meeting, however HPPTCL views are noted.

Considering the implementation schedule of Shongtong HEP (Nov'26) following interim arrangement is agreed to be implemented by HPPTCL for evacuation of power from Shongtong HEP :

- LILO of one ckt of 400kV Karcham Wangtoo- Baspa D/c line (Triple snowbird) at generation switchyard of Shongtong HEP (Quad$) *

* ***After implementation of final scheme, above LILO portion (as part of interim arrangement) will be opened and part utilized for LILO of one circuit of Jhangi PS - Wangtoo (HPPTCL) 400 kV D/c (Quad) line at generation switchyard of Shongtong HEP as part of final scheme***.

$ Line capacity shall be 2500 MVA per circuit at nominal voltage

 # The matter was deliberated in the 27th NRPC TeST held on 21.04.2025 wherein HPPTCL confirmed that OPGW is considered on the LILO portion of 400kV Karcham Wangtoo- Baspa D/c line at Shongtong HEP

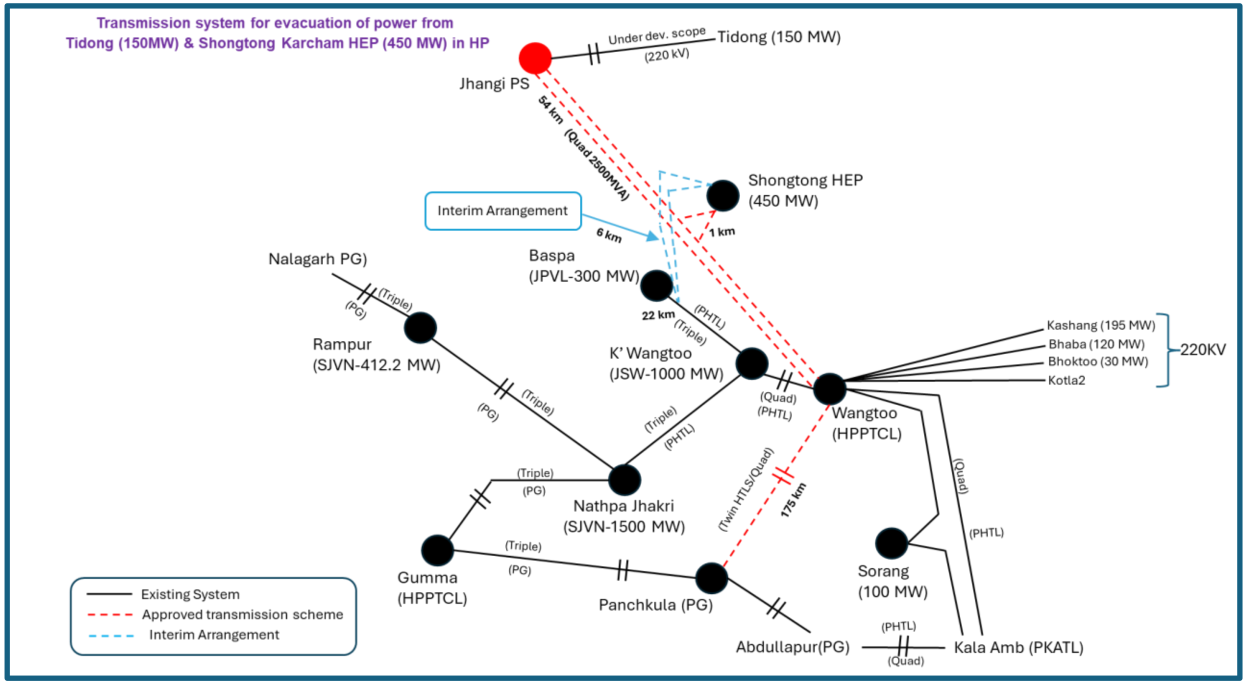


Fig: Schematic diagram of Comprehensive transmission scheme for evacuation of Tidong and Shongtong HEP

C2. Implementation of 400/220kV (2x500MVA), substation, Khurpia Farm, Udham Singh Nagar, Uttarakhand

It was stated that PTCUL vide letter dated 28.11.2024 submitted agenda for Construction of 400/220/132/33 (2X500MVA+2X160MVA+2X80MVA), substation, Khurpia Farm, Udham Singh Nagar, Uttarakhand **.** In this agenda PTCUL informed that Government of India, is developing various Industrial Corridor Projects as part of National Industrial Corridor program. The Union Commerce and Industry Minister during a virtual press conference announced that the Khurpiya farm in Kichha located in the Udhamsingh Nagar district, Uttarakhand is now to be transformed into a smart industrial city under the Amritsar Kolkata Industrial Corridor (AKIC). The initiative marks a pivotal moment in the industrial development of Uttarakhand, especially with its focus on establishing Khurpiya as an automobile hub, alongside industries related to engineering and fabrication. Further Presently there is transmission constraints during peak load on 220 kV Pantnagar-Bareilly (UPPTCL) line.

PTCUL also mentioned that State Industrial Development Corporation of Uttarakhand Ltd. (SIDCUL) vide letter No. 69/MD/SIDCUL/2022 dated 23-07-2022 requested ultimate power requirement of 142 MVA for the earlier proposed Industrial Estate. Further to meet the power demand of upcoming AIIMS, Modern Degree College, Roadways Bus Stand near Khurpiya Farm, smart industrial city & To cater increasing load demand of Kumaon region & strengthening of transmission system in Pantnagar and nearby area and to provide reliable power supply, 400/220/132/33 Khurpia Farm Substation is required.

First Phase: Substation & Associated Transmission lines

- 220/132/33 kV Substation, Khurpia Farm (2 X 160 MVA T/F, 220/132 kV and 2X80 MVA, T/F 132/33 kV)
- LILO of 220KV Bareilly-Pantnagar Line at proposed Substation Khurpiya Farm. (4KM)
- LILO of 132KV Kichha-Rurdrapur line at proposed Substation Khurpiya Farm(4 KM)

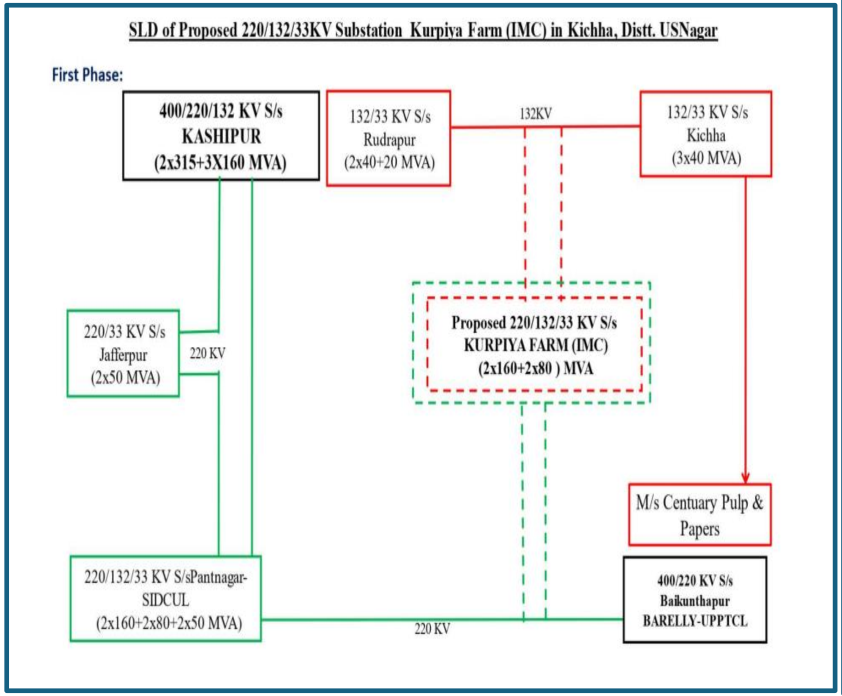
Completion time Frame: commissioning of phase-I by 31.12.2027

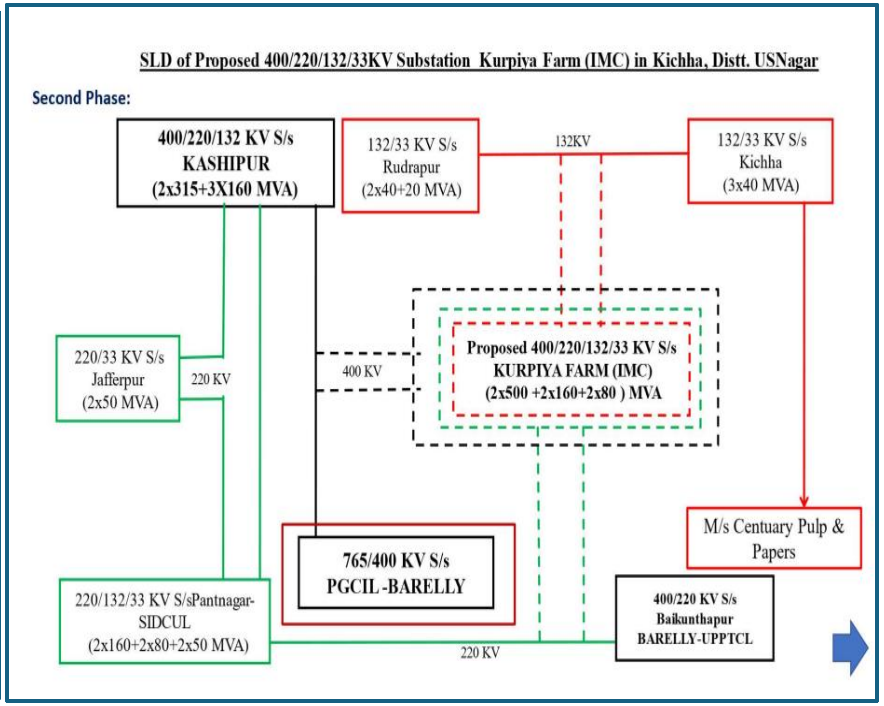
Second Phase: Substation & Associated Transmission lines

- 400/220 kV Substation, Khurpia Farm (2 X 500 MVA)
- LILO of 400 kV Bareilly- Kashipur line (24KM)

Completion time Frame: commissioning of phase-II by 31.12.2029

PTCUL in the letter mentioned that in present condition PTCUL can cater only 50 MVA load demand by construction of 220/132/33 KV Substation in 1st Phase due to load restriction on 220 kV Pantnagar-Bareilly (UPPTCL) line and rest of the load demand can only be met after construction of 400/220KV Substation in 2nd Phase. Total proposed load of 142 MVA of SIDCUL will be required in phased manner. In first phase some important loads like AIIMS etc. (around 40 to 50 MW) need to be supplied in next 02 years so first phase has been proposed. Further 400 kV S/s will be required to meet ultimate load demand of SIDCUL (142 MVA) which may further increased in future.





Further PTCUL mentioned that construction of 400/220/132/33 kV Substation, (2x500MVA+2x160MVA+2x80MVA), Khurpia Farm will reduce the overloading problem of 400 kV S/s Kashipur & creates additional source of power for 220 kV S/s Pantnagar, 132 kV S/s Kichha and 132 kV S/s Rudrapur. Considering the above, PTCUL has proposed the construction of 400/220/132/33 kV Substation, (2x500MVA+2x160MVA+2x80MVA), Khurpia Farm, Udham Singh Nagar, Uttarakhand with associated lines to meet the proposed load demand of SIDCUL (142 MVA), future load growth of nearby border areas and for strengthening the transmission network in hills of Kumaun region.

CTUIL vide mail dated 06.12.24 to CEA, Grid-India & UP requested to provide comments/observations on PTCUL's Khurpia Farm S/s proposal at the earliest for taken up in ensuing CMETS-NR meeting. Grid-India vide mail dated 17.12.2024 given following comments/observations on PTCUL's proposal:

- As per recent loading pattern in the area, it is to be noted that there are severe high loading issues in the Kashipur area. 400/220kV Kashipur ICTs are N-1 non-compliant especially during summer months when gas generation at Shravanti and Gamma Infra is not available. There have been number of tripping already in Kashipur area during tripping of 220kV Bareilly-Pantnagar line. Subsequently, SPS has been implemented in the complex to avoid major load loss.

- Further, in case LILO of 220KV Bareilly-Pantnagar Line is implemented at Khurpiya Farm, the issue will aggravate. As such it is important that generation at Shravanti and Gamma Infra is available in case additional power is to be drawn from 220kV Bareilly-Pantnagar line. Thus, it is important that additional load is provided in matching timeframe of 400/220kV Khurpia Farm S/s.

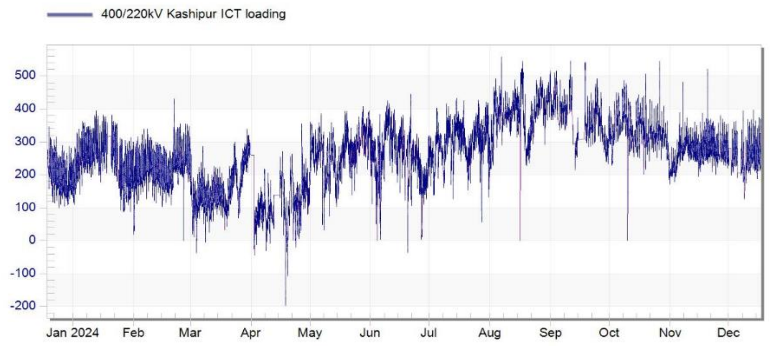


Fig : Loading of Kasipur ICTs (2x315MVA) (Source: Grid-India)

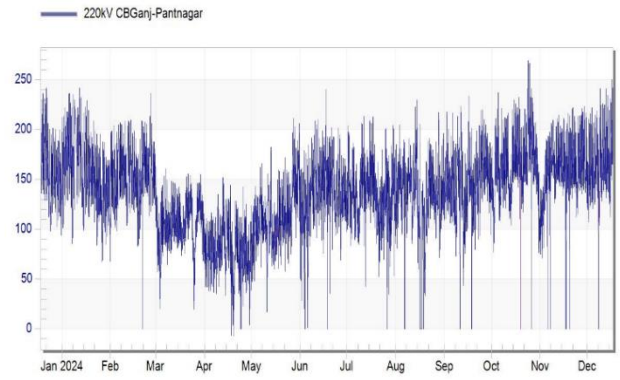


Fig : Loading of 220kV CB Ganj-Pantnagar line of Past one year (Source: Grid-India)

As per CTU studies (2027-28 timeframe), 400/220kV Kashipur ICTs are N-1 non-compliant in summer season & loadings further increase when gas generation at Shravanti and Gamma Infra is not available. From studies it emerged that considering Sravanthi Generation (3x75MW (80% dispatch)) & Gama generation (2x75MW (80% dispatch)) & without considering PTCUL proposal , loading of Kashipur ICT is 2x285MW (N-1 loading 437MW). Loading of Kashipur ICT further increases to 373MW/ICT (2x315MVA) when gas generation at Shravanti and Gamma Infra is not available.

Further with PTCUL proposed Phase-1 scheme (considering load of Ph-I at Khurpiya farm & Sravanthi Generation (3x75MW) & Gama generation (2x75MW)), Loading of Kashipur ICT increases by ~15MW/ICT from base case, However with proposed Phase-1 & Phase-2 combined scheme (considering total load of 140MW at Khurpiya farm & Sravanthi Generation (3x75MW) & Gama generation (2x75MW)), Loading of Kashipur ICT reduces by ~85MW/ICT (199MW/ICT(N-1 loading-275MW)). Therefore, it is recommended that proposed Phase-1 & Phase-2 scheme may be implemented in same timeframe, matching with the augmentation of Kashipur ICT (400/220kV, 3rd ICT). PTCUL may update the status of augmentation of 400/220kV ICT at Kashipur S/s.

Further considering Phase-1 & Phase-2 combined scheme, loading on 220kV Khurpiya Farm-Pantnagar line (formed after LILO) is on higher side (~235MW under normal condition), when gas generation at Shravanti and Gamma Infra is not available. CEA and NRLDC may provide their inputs on high loading and requirement of reconductoring of above line/reconfiguration of feeders to control loading on 220kV Khurpiya Farm-Pantnagar line.

PTCUL vide mail dated 15.02.2025 requested to take up the proposal in upcoming meeting due to urgency of 400/220/132 kV substation in Khurpiya Farm for development of Nation Industrial corridor development and smart industrial city in khurpiya farm, Kichha. PTCUL also stated that they are facing enormous pressure from state administration to initiate tendering process at the earliest. PTCUL in the mail informed that they are agreeing to construct the 400/220/132kV substation instead 220kV substation in first phase and 400/220 kV substation in second phase. Further CTUIL vide mail dated 19.02.2025, requested PTCUL to provide the implementation time frame for the proposed Intra state transmission scheme. PTCUL vide mail 06.03.25 stated that subject Intra state transmission scheme is as under considering the award date is 01.07.2025

- 400/220/132kV/33kV S/s, Khurpia Farm (2 X 500 MVA, 400/220kV; 2 X 160 MVA T/F, 220/132 kV and 2X80 MVA, T/F 132/33 kV)) (implementation time =3.5 years)
- LILO of 400 kV Bareilly- Kashipur line at Khurpia Farm (24KM) (implementation time =3.5 years due to propable RoW issues)
- LILO of 220KV Bareilly-Pantnagar Line at proposed S/s Khurpiya Farm (4KM) (implementation time =1 years)
- LILO of 132KV Kichha-Rurdrapur line at proposed S/s Khurpiya Farm(4 KM) (implementation time =1 years)

Implementation time Frame: commissioning 400/220/132/33kV substation by 31.12.2028

In the meeting, CTU enquired about implementation timeframe of various elements of proposed intra state transmission scheme. PTCUL stated that complete substation i.e. 400/220/132/33kV Khurpia farm will be awarded in implementation timeframe of Dec'28 and implementation of 220kV and 132kV lines will be implemented in matching timeframe of 400kv system considering comment received from CTU and Grid-India.

CTU & NRLDC further reiterated that considering the Grid constraint, it is prudent that all the elements of scheme may be implemented in same timeframe and augmentation of Kashipur ICT (400/220kV, 3rd ICT) shall be expedite by PTCUL so that ICT would be available on or before implementation of proposed PTCUL scheme. PTCUL stated that for Kashipur ICT augmentation, tender is already opened and ICT will be available before Dec'28.

Considering above following intra state scheme is agreed to be implemented by PTCUL:

- LILO of 220KV Bareilly-Pantnagar Line at proposed S/s Khurpiya Farm (4KM)
- LILO of 400 kV Bareilly- Kashipur line at Khurpia Farm (24km)

Implementation time Frame: 31.12.2028 (matching with commissioning of 400/220/132/33kV S/s along with LILO of 400 kV Bareilly- Kashipur line at Khurpia Farm)

CTU stated that establishment of 400/220/132/33 kV Khurpia Farm Substation and LILO of 132KV Kichha-Rurdrapur line at proposed Substation Khurpiya Farm is in intra state in nature therefore, PTCUL is requested to taken up approval through CEA for above part of scheme. CTU requested to CEA to provide approval of above proposed intra state scheme to PTCUL based on deliberation in present meeting.

C3. 220 kV Sectionalizer bay (1 set), 220 kV BC (1 nos.) bays and 220 kV TBC (1 nos.) bay at Bikaner-IV PS

It was stated that augmentation of 2x500 MVA (7th & 8th), 400/220 kV ICTs at Bikaner-IV PS was discussed and agreed in 36th CMETS-NR meeting held on 15/01/2025. Considering above ICT augmentation as well as no. of 220kV line bays for RE interconnection, 220 kV Sectionalizer bay, 220 kV BC bay and 220 kV TBC bay at Bikaner-IV PS is required. No comments were received from stakeholders.

In view of above, following scheme is agreed in ISTS:

- 220 kV Sectionalizer bay (1 set), 220 kV BC (1 no.) bay and 220 kV TBC (1 no.) bay at Bikaner-IV PS

C4. Augmentation with 400/220 kV 1x500 MVA (4th) ICT at Sikar (PG) S/s to meet N-1 compliance

It was stated that in the 225th OCC meeting held on 12.11.24, NRLDC agenda for N-1 non-compliance of 400/220kV ICT at Sikar (PG) Substation was discussed. In the OCC meeting CTUIL was asked to take up ICT augmentation of 400/220kV Neemrana, Sikar and Jaipur South POWERGRID substations in upcoming CMETS meeting. The ICTs at Neemrana & Jaipur(South) are already discussed and agreed in 36th CMETS-NR meeting, however for Sikar ICT, agenda was not taken up due to pending space availability date from POWERGRID.

At present, 400/220kV Sikar (PG) Substation has total transformation capacity of 1130 MVA (1x500MVA+2x315MVA). From the loading pattern of Sikar ICTs (1x500 MVA+2x315MVA), it is observed that cumulative maximum loading on 400/220 kV ICTs is about 950 MW. From the analysis, it emerged that loading is breaching N-1 limit (820 MW) in summer season (Jun’24-Jul’24) for sufficient duration of time. The loading pattern of Sikar ICTs for past one year is as below :

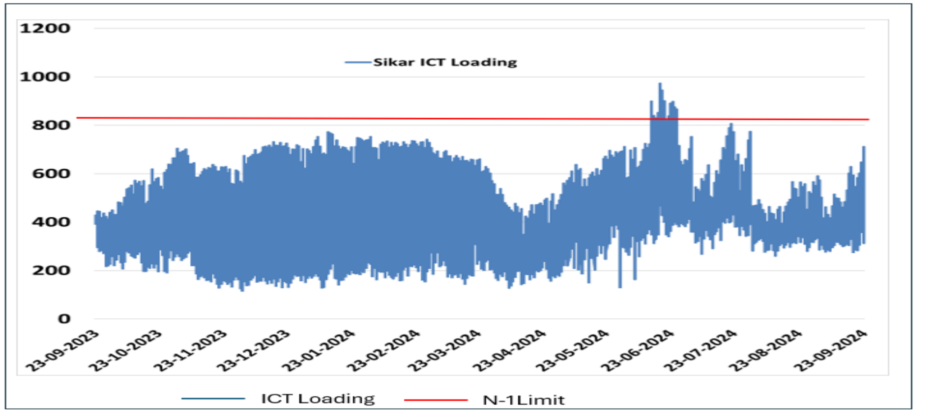


Fig : Loading of Sikar ICTs of Past one year (Source: Grid-India)

POWERGRID vide mail dated 15.01.2025 confirmed space availability for Augmentation of 400/220 kV transformer (4^{th}, 500MVA) along with transformer bays (with 220kV side connection with cable) at 400/220kV Sikar (PG) S/s.

In the meeting, Grid-India stated that as discussed in OCC/NRPC meeting as well as highlighted in operational feedback, there is augmentation requirement at Sikar(PG) S/s considering n-1 non-compliance issue. In the previous winters (2024) season, loadings were higher w.r.t loadings in last year (as depicted in figure). CTU stated that RVPN may confirm the requirement for ICT augmentation (4^{th}, 1x500MVA) at 400/220kV Sikar (PG) S/s.

RVPN stated that at present, no additional interconnections are planned at Sikar (PG) S/s for drawl of power and RVPN already planned Beri Bajranggarh S/s to divert partial load of about 125MW from Sikar (PG) S/s to their Beri Bajranggarh S/s which is ultimately fed through Ratangarh S/s. The scheme is already awarded with implementation schedule of 24 months.

Powergrid stated that 800MW-900MW average load on three ICTs and in case of outage of 500MVA ICT, other two ICTs of 315MVA will become N-1 non-compliant. RVPN stated that at 400kV Ratangarh S/s, additional ICT of 500MVA is recently approved which will also cater the load of same area. RVPN also stated that their Jaipur (North) S/s is under planning which will also share the load of Reengus area which is near to same complex.

RVPN requested that they will carry out detailed analysis for the requirement of ICT augmentation at Sikar (PG) S/s based on under implementation and planned intra state system envisaged in next 2-3 years. RVPN requested forum to take up the agenda in next CMETS-NR meeting.

CTUIL enquired to POWERGRID about commissioning year of 315MVA ICTs for analyse the feasibility of replacement of 315MVA ICTs with 500MVA. Powergrid replied that 2 nos. of 315MVA ICTs were commissioned in 2012. Considering commissioning of old 315MVA transformers, forum decided that augmentation with new 500MVA (4th) ICT is to be considered based on RVPN study analysis and agenda will be discussed in next CMETS-NR meeting.

Annexure-I

Transmission system for Connectivity under GNA at Barmer-III PS(Tentative)

1. Establishment of 400kV Pooling Station at suitable location near Barmer (Barmer-III PS) along with 2x125 MVAr bus reactor
2. 400/220kV 7x500MVA ICTs at Barmer-III PS (for connectivity agreed for grant at 220kV level)
3. LILO of both ckts of 400kV Barmer-I PS – Barmer-II PS D/c line (Quad) at Barmer-III PS
4. Establishment of 6000 MW, Barmer-III(HVDC) terminal station (4x1500 MW) at a suitable location near Barmer-III PS
5. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
6. HVDC line between Barmer-III (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
7. Associated EHVAC system strengthening in WR/SR/ER
8. 2x[+]300/(-)210MVAr Synchronous condenser at 400kV level of Barmer-III PS

Annexure-II

Transmission system for Connectivity under GNA at Bhadla-V PS(Tentative)

1. Establishment of 765/400kV, 2x1500 MVA pooling station at suitable location near Bhadla (Bhadla-V PS) along with 2x125 MVAr & 2x240 MVAr bus reactor along with 2x125 MVAr & 2x240 MVAr bus reactor
2. 400/220kV 2x500MVA ICTs at Bhadla-V PS (for connectivity agreed for grant at 220kV level)
3. Bhadla-IV PS – Bhadla-V PS 400kV D/c line (Quad)
4. 765kV Bhadla-V PS – Bikaner-V PS D/c line
5. Establishment of 6000 MW, ± 800 kV Bhadla-V (HVDC) terminal station (4x1500 MW) at suitable location near Bhadla
6. Establishment of 6000 MW, ±800 kV terminal station (4x1500 MW) at suitable location in WR/ER/SR
7. ±800 kV HVDC line between Bhadla-V (HVDC) & Suitable location in WR/ER (with Dedicated Metallic Return)
8. Establishment of 765/400kV, 2x1500 MVA S/s substation station at suitable location in WR/ER/SR along with 2x125 MVAr & 2x240 MVAr bus reactor
9. Associated transmission system in WR/ER for onwards dispersal of power

Annexure-III

Transmission system for Connectivity under GNA at Ramgarh-II PS (Tentative)

1. Establishment of 2x1500 MVA, 765/400 kV Ramgarh-II Pooling Station at a suitable location near Ramgarh along with 2x125 MVAr & 2x240 MVAr bus reactor
2. Ramgarh-II PS – Bhadla-IV PS 765 kV D/c line along with 240 MVAr switchable line reactor for each circuit at Ramgarh-II PS end
3. Ramgarh PS – Ramgarh-II PS 400 kV D/c line (Quad)
4. LILO of both ckts of 765kV Ramgarh PS – Bhadla-III PS D/c line at Ramgarh-II PS along with 240 MVAr switchable line reactor for each circuit at Ramgarh-II PS end
5. Establishment of 6000 MW, Ramgarh(HVDC) terminal station (4x1500 MW) at a suitable location near Ramgarh-II substation
6. Establishment of 6000 MW, HVDC terminal station (4x1500 MW) at suitable location in WR/SR/ER
7. HVDC line between Ramgarh (HVDC) & HVDC terminal station in WR/SR/ER (with Dedicated Metallic Return)
8. Associated EHVAC system strengthening in WR/SR/ER
9. 2x[+]300/(-)210MVAr Synchronous condenser at 400kV level of Ramgarh-II PS

Additional system for connectivity at 220kV level only of Ramgarh-II PS

1. 400/220kV, 3x500 MVA ICTs at Ramgarh-II PS

Annexure-A

List of Participants of 37th Consultation meeting for Evolving Transmission Schemes in NR held on 27.03.2025

CEA

Shri Nitin Deswal	Deputy Director

SECI

Shri Vineet Kumar	DGM
Shri R. K. Agarwal	Consultant

Grid India

Shri Gaurav Malviya	Manager, NRLDC

CTUIL

Shri Partha Sarathi Das	Sr. GM (CTUIL)
Shri Sandeep Kumawat	DGM (CTUIL)
Shri V. M. S. Prakash Yerubandi	DGM (CTUIL)
Smt Ankita Singh	DGM (CTUIL)
Shri Narendra Sathvik Ranganath	Ch. Manager (CTUIL)
Shri Yatin Sharma	Manager (CTUIL)
Shri Madhusudan Meena	Engineer (CTUIL)
Shri Rishabh Bansal	ET (CTUIL)
Shri Kumar Anjul	ET (CTUIL)
Shri Jayesh Raikwar	ET (CTUIL)
Shri Avinash Meena	ET (CTUIL)

POWERGRID

Shri Shafat Ahmad Wani	Sr. GM

Shri Vijay Srivastava	Sr. DGM
Shri Rakesh Kumar Gupta	DGM
Shri Yashpal Choudhary	DGM
Shri Rohit Kumar Jain	DGM
Shri Hukmichand Menaria	DGM

JKPTCL

Shri Ajaz Ahmad Sheikh	SE
Mohd. Amin	Technical Officer

RVPNL

Dr. Om Prakash Mahela	XEN(PP&D)

PTCUL

Shri G.S. Budiyal	Director (Operations)
Shri Rajeev Gupta	CE(L-1)
Shri D.C. Pande	CE
Shri Rakesh Kumar	SE
Smt Saima Kamal	SE
Shri Ashok Kumar	Ex. En.

HPPTCL

Shri Harmanjeet Singh	AE(Plg.)

HVPN

Shri Sanjay Verma	SE(Plg.)

UPPTCL

Shri Satyendra Kumar	SE

Connectivity/GNA Applicants

Shri Vaibhav Kapoor	Aditya Birla Renewables Limited
Shri Jayakumar Pillai	Aditya Birla Renewables Limited
Shri Gopal Eti	Apraava Energy Private Limited
Shri Manjunath Hidakal	Apraava Energy Private Limited
Shri Angshuman Rudra	Avaada Rjbikaner/Rjsustainable/Energy Private Limited
Shri Vishal Kumar	Azure Power Fifty One Private Limited
Shri Harshit Gupta	Hexa Climate Solutions Private Limited
Shri Mahesh K. Mandwarya	Jubilant Ingrevia Limited
Shri C. B. Bhardwaj	Jubilant Ingrevia Limited
Shri Puneet Gupta	Jubilant Ingrevia Limited
Shri Soni kumar	SJVNL
Shri Prakash Singh	SJVNL
Shri Poorva Pitke	Sprng Green Energy 4 Private Limited
Shri Chinmay Sirdeshmukh	Sprng Green Energy 4 Private Limited
Shri Santosh Narayan	Tata Power Renewable Energy Limited
Shri Mahadev S Udachan	Tata Power Renewable Energy Limited
Shri K. A. Vishwanath	TEQ Green Power IX/XIII Private Limited
Shri Varun Bhatnagar	Unique Hytech Renewable Energy Private Limited
Shri Aniket Vyas	SA Renewable Infra Private Limited
Shri Naveen Singh	Pokhran Wind Energy Private Limited
Shri Mohit Jain	Renew Solar Power Private Limited
Shri Aditya Thakur	HPPCL